

### 2.1.4 Exercises

In the following exercises, feel free to use what you know from calculus to find the limit, if it exists. But you must *prove* that you found the correct limit, or prove that the sequence is divergent.

**Exercise 2.1.1:** Is the sequence  $\{3n\}_{n=1}^{\infty}$  bounded? Prove or disprove.

**Exercise 2.1.2:** Is the sequence  $\{n\}_{n=1}^{\infty}$  convergent? If so, what is the limit?

**Exercise 2.1.3:** Is the sequence  $\left\{\frac{(-1)^n}{2n}\right\}_{n=1}^{\infty}$  convergent? If so, what is the limit?

**Exercise 2.1.4:** Is the sequence  $\{2^{-n}\}_{n=1}^{\infty}$  convergent? If so, what is the limit?

**Exercise 2.1.5:** Is the sequence  $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$  convergent? If so, what is the limit?

**Exercise 2.1.6:** Is the sequence  $\left\{\frac{n}{n^2+1}\right\}_{n=1}^{\infty}$  convergent? If so, what is the limit?

**Exercise 2.1.7:** Let  $\{x_n\}_{n=1}^{\infty}$  be a sequence.

a) Show that  $\lim_{n \rightarrow \infty} x_n = 0$  (that is, the limit exists and is zero) if and only if  $\lim_{n \rightarrow \infty} |x_n| = 0$ .

b) Find an example such that  $\{|x_n|\}_{n=1}^{\infty}$  converges and  $\{x_n\}_{n=1}^{\infty}$  diverges.

**Exercise 2.1.8:** Is the sequence  $\left\{\frac{2^n}{n!}\right\}_{n=1}^{\infty}$  convergent? If so, what is the limit?

**Exercise 2.1.9:** Show that the sequence  $\left\{\frac{1}{\sqrt[3]{n}}\right\}_{n=1}^{\infty}$  is monotone and bounded. Then use [Theorem 2.1.10](#) to find the limit.

**Exercise 2.1.10:** Show that the sequence  $\left\{\frac{n+1}{n}\right\}_{n=1}^{\infty}$  is monotone and bounded. Then use [Theorem 2.1.10](#) to find the limit.

**Exercise 2.1.11:** Finish the proof of [Theorem 2.1.10](#) for monotone decreasing sequences.

**Exercise 2.1.12:** Prove [Proposition 2.1.13](#).

**Exercise 2.1.13:** Let  $\{x_n\}_{n=1}^{\infty}$  be a convergent monotone sequence. Suppose there exists a  $k \in \mathbb{N}$  such that

$$\lim_{n \rightarrow \infty} x_n = x_k.$$

Show that  $x_n = x_k$  for all  $n \geq k$ .

**Exercise 2.1.14:** Find a convergent subsequence of the sequence  $\{(-1)^n\}_{n=1}^{\infty}$ .

**Exercise 2.1.15:** Let  $\{x_n\}_{n=1}^{\infty}$  be a sequence defined by

$$x_n := \begin{cases} n & \text{if } n \text{ is odd,} \\ 1/n & \text{if } n \text{ is even.} \end{cases}$$

a) Is the sequence bounded? (prove or disprove)

b) Is there a convergent subsequence? If so, find it.

**Exercise 2.1.16:** Let  $\{x_n\}_{n=1}^{\infty}$  be a sequence. Suppose there are two convergent subsequences  $\{x_{n_i}\}_{i=1}^{\infty}$  and  $\{x_{m_i}\}_{i=1}^{\infty}$ . Suppose

$$\lim_{i \rightarrow \infty} x_{n_i} = a \quad \text{and} \quad \lim_{i \rightarrow \infty} x_{m_i} = b,$$

where  $a \neq b$ . Prove that  $\{x_n\}_{n=1}^{\infty}$  is not convergent, without using [Proposition 2.1.17](#).

**Exercise 2.1.17 (Tricky):** Find a sequence  $\{x_n\}_{n=1}^{\infty}$  such that for every  $y \in \mathbb{R}$ , there exists a subsequence  $\{x_{n_i}\}_{i=1}^{\infty}$  converging to  $y$ .

**Exercise 2.1.18 (Easy):** Let  $\{x_n\}_{n=1}^{\infty}$  be a sequence and  $x \in \mathbb{R}$ . Suppose for every  $\epsilon > 0$ , there is an  $M$  such that  $|x_n - x| \leq \epsilon$  for all  $n \geq M$ . Show that  $\lim_{n \rightarrow \infty} x_n = x$ .

**Exercise 2.1.19 (Easy):** Let  $\{x_n\}_{n=1}^{\infty}$  be a sequence and  $x \in \mathbb{R}$  such that there exists a  $k \in \mathbb{N}$  such that for all  $n \geq k$ ,  $x_n = x$ . Prove that  $\{x_n\}_{n=1}^{\infty}$  converges to  $x$ .

**Exercise 2.1.20:** Let  $\{x_n\}_{n=1}^{\infty}$  be a sequence and define a sequence  $\{y_n\}_{n=1}^{\infty}$  by  $y_{2k} := x_{k^2}$  and  $y_{2k-1} := x_k$  for all  $k \in \mathbb{N}$ . Prove that  $\{x_n\}_{n=1}^{\infty}$  converges if and only if  $\{y_n\}_{n=1}^{\infty}$  converges. Furthermore, prove that if they converge, then  $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n$ .

**Exercise 2.1.21:** Show that the 3-tail of the sequence defined by  $x_n := \frac{n}{n^2+16}$  is monotone decreasing. Hint: Suppose  $n \geq m \geq 4$  and consider the numerator of the expression  $x_n - x_m$ .

**Exercise 2.1.22:** Suppose that  $\{x_n\}_{n=1}^{\infty}$  is a sequence such that the subsequences  $\{x_{2n}\}_{n=1}^{\infty}$ ,  $\{x_{2n-1}\}_{n=1}^{\infty}$ , and  $\{x_{3n}\}_{n=1}^{\infty}$  all converge. Show that  $\{x_n\}_{n=1}^{\infty}$  is convergent.

**Exercise 2.1.23:** Suppose that  $\{x_n\}_{n=1}^{\infty}$  is a monotone increasing sequence that has a convergent subsequence. Show that  $\{x_n\}_{n=1}^{\infty}$  is convergent. Note: So [Proposition 2.1.17](#) is an “if and only if” for monotone sequences.